

Derivatives of elementary functions



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Derivatives of elementary functions

- In what follows we derive the derivative functions of some of the main elementary functions.

Remark 1

Given the function $f(x) = c$ with $c \in \mathbb{R}$ constant, the derivative function is $f'(x) = 0$.

Proof. We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 0 = 0$$

□

1.1 Derivatives of power functions

Remark 2

Given the function $f(x) = x^n$ with $n \in \mathbb{N}, n \geq 1$, the derivative function is $f'(x) = n x^{n-1}$.

Proof. From the binomial theorem (Newton's binomial formula) we have:

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^n - x^n}{h} = \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\ &= \frac{x^n + n x^{n-1} h + \sum_{k=2}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\ &= n x^{n-1} + \frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h} \end{aligned}$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0} = n x^{n-1}$$

□

Proof. With $n = 2$, we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$$

□

Remark 3

Given the function $f(x) = x^\alpha$ with $\alpha \in \mathbb{R}$, the derivative function is $f'(x) = \alpha x^{\alpha-1}$ for $x > 0$.

Proof. Let $x > 0$. We have:

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^\alpha - x^\alpha}{h} = \frac{\left(x\left(1 + \frac{h}{x}\right)\right)^\alpha - x^\alpha}{h} \\ &= x^\alpha \cdot \frac{\left(1 + \frac{h}{x}\right)^\alpha - 1}{h} \sim x^\alpha \cdot \frac{\alpha \frac{h}{x}}{h} = \alpha x^{\alpha-1} \text{ for } h \rightarrow 0 \end{aligned}$$

where we used the fundamental limit

$$(1 + \varepsilon(h))^\alpha - 1 \sim \alpha \varepsilon(h) \text{ for } \varepsilon(h) \rightarrow 0$$

where

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \text{ for } h \rightarrow 0.$$

Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \alpha x^{\alpha-1} = \alpha x^{\alpha-1}$$

□

Example 1: Derivative function

$$f(x) = x^{10} \qquad f'(x) = 10x^9 \quad (\text{for } x \in \mathbb{R})$$

$$f(x) = \frac{1}{x} = x^{-1} \qquad f'(x) = -\frac{1}{x^2} \quad (\text{for } x > 0)$$

$$f(x) = \sqrt{x} = x^{\frac{1}{2}} \qquad f'(x) = \frac{1}{2\sqrt{x}} \quad (\text{for } x > 0)$$

1.2 Derivatives of elementary trigonometric functions

Remark 4

Given the function $f(x) = \sin x$, the derivative function is $f'(x) = \cos x$.

Proof. Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\sin(x+h) - \sin x}{h} = \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} = \\ &= \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x\end{aligned}$$

Using the fundamental limit

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2} \text{ for } h \rightarrow 0$$

we have

$$\frac{\cos h - 1}{h} = h \cdot \left(\underbrace{-\frac{1 - \cos h}{h^2}}_{\rightarrow -\frac{1}{2} \text{ for } h \rightarrow 0} \right) \sim -\frac{1}{2} h \text{ for } h \rightarrow 0.$$

Using the fundamental limit

$$\frac{\sin h}{h} \rightarrow 1 \text{ for } h \rightarrow 0$$

we have

$$\frac{\sin h}{h} \cos x \sim \cos x \text{ for } h \rightarrow 0.$$

Then

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x \\ &= \lim_{h \rightarrow 0} \sin x \left(-\frac{1}{2} h \right) + \cos x = \cos x\end{aligned}$$

□

Remark 5

Given the function $f(x) = \cos x$, the derivative function is $f'(x) = -\sin x$.

Proof. Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\cos(x+h) - \cos x}{h} = \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \\ &= \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x\end{aligned}$$

and, using the arguments of the previous proof, we have:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x \\ &= \lim_{h \rightarrow 0} \cos x \left(-\frac{1}{2} h \right) - \sin x = -\sin x\end{aligned}$$

□

1.3 Derivatives of the exponential and logarithmic functions with base e

Remark 6

Given the function $f(x) = e^x$, the derivative function is $f'(x) = e^x$.

Proof. We have

$$\frac{f(x+h) - f(x)}{h} = \frac{e^{x+h} - e^x}{h} = e^x \cdot \frac{e^h - 1}{h} \sim e^x \text{ for } h \rightarrow 0$$

using the fundamental limit

$$\frac{e^h - 1}{h} \rightarrow 1 \text{ for } h \rightarrow 0.$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} e^x = e^x$$

□

Remark 7

Given the function $f(x) = \log x$, the derivative function is $f'(x) = \frac{1}{x}$.

Proof. We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{\log(x+h) - \log x}{h} = \frac{\log\left(1 + \frac{h}{x}\right)}{h} \sim \frac{h}{x} \cdot \frac{1}{h} = \frac{1}{x} \text{ for } h \rightarrow 0$$

where we used the fundamental limit

$$\log(1 + \varepsilon(h)) \sim \varepsilon(h) \text{ for } \varepsilon(h) \rightarrow 0,$$

and

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \text{ for } h \rightarrow 0.$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{x} = \frac{1}{x}$$

□

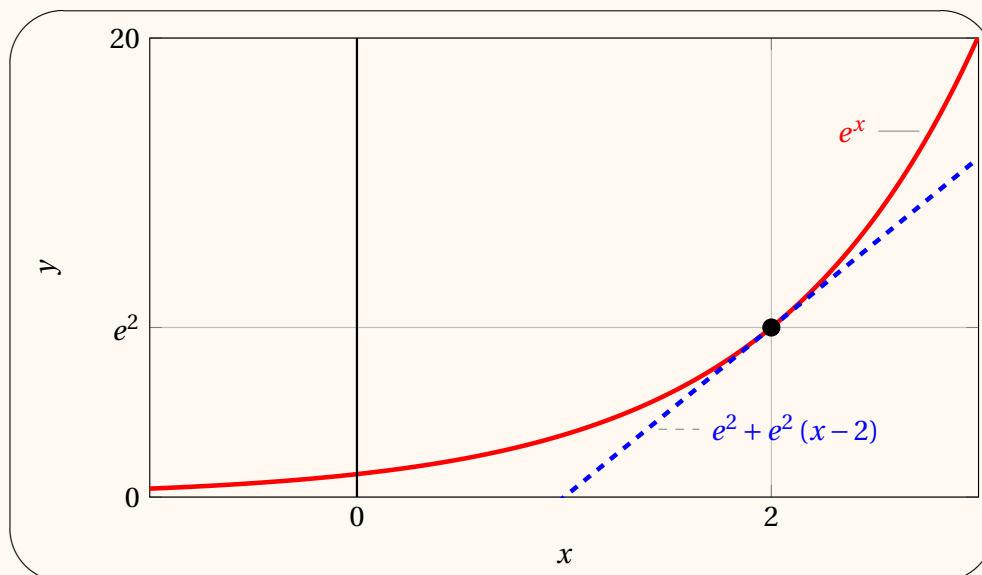
Example 2: Tangent line

We compute the equation of the tangent line to the graph of the function

$$f(x) = e^x \text{ at the point with abscissa } x = 2$$

We have $f(2) = e^2$, $f'(x) = e^x$, $f'(2) = e^2$, hence the tangent line at the point $(2, e^2)$ is:

$$y = f(2) + f'(2)(x-2) = e^2 + e^2(x-2)$$



Example 3: Tangent line

We compute the equation of the tangent line to the graph of the function

$$f(x) = x^3 \text{ at the point with abscissa } x = 2$$

We have $f(2) = 8$, $f'(x) = 3x^2$, $f'(2) = 12$, hence the tangent line at the point $(2, 8)$ is:

$$y = f(2) + f'(2)(x - 2) = 8 + 12(x - 2)$$

